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Clamping system, balancing system and methods of use thereof

Abstract

A clamping system for a planetary ball mill includes a base, a clamping lid, including a clamping mechanism configured to secure a grinding container between the base and the clamping mechanism, and a lid configured to pivot with respect to the base from a closed position to an open position, wherein in the open position the grinding container is removable from the clamping system. The system further includes a rotatable arm configured to pivot between a first and second position, wherein in the first position the rotatable arm prevents the clamping lid from rotation to the open position. The clamping system also includes a balancing system having a first counterweight, with a linkage system being coupled between the first counterweight and a carrier plate of the planetary ball mill, the linkage system allowing the first counterweight to move with respect to a rotational axis of the carrier plate.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to, and the benefit of, U.S. Provisional Application No. 63/219,565, Titled: “Clamping System, Balancing System and Methods of Use Thereof,” filed on Jul. 8, 2021, the entirety of the contents of the preceding application is incorporated herein by reference, as if fully set forth in this document, for all purposes.

INTRODUCTION

(1) The present disclosure generally relates to ball mills, particularly to a ball mill clamping system

and a balancing system for planetary ball mills.

BACKGROUND

(2) A ball mill may be used to grind a material into powder. The ball mill may include a grinding jar containing a plurality of grinding balls and a material to be ground. The ball mill typically rotates the grinding container about a longitudinal axis of the grinding container. As the grinding container rotates, the grinding balls roll along the inside of the grinding container and then fall onto the material to be ground, grinding the material to be ground into a fine powder.

(3) Conventional ball mills may include clamping mechanisms in an attempt to clamp the container to the rotating ball mill. Conventional ball mill grinding container clamps frequently require a series of threaded shafts and nuts and a plate to sandwich the grinding container between two surfaces. Such clamping mechanisms are inefficient and tedious to install and remove. Further, in a conventional clamping mechanism it may be difficult to ensure that the grinding container is properly secured to the ball mill. Thus, the need exists to provide an improved clamping system and grinding container. In addition, conventional ball mills often have similarly difficult or tedious balancing procedures. Thus, there further exists a need to improve efficiency in balancing a ball mill.

SUMMARY

(4) The following presents a simplified summary of one or more aspects of the invention in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(5) In one aspect, the disclosure, a clamping system for a planetary ball mill is disclosed. The clamping system includes a base and a clamping lid including a clamping mechanism configured to secure a grinding container between the base and the clamping mechanism. The clamping lid is configured to pivot with respect to the base from a closed position to an open position, wherein in the open position the grinding container is removable from the clamping system. The system further includes a rotatable arm configured to pivot from a first position to a second position, wherein in the first position the rotatable arm prevents the clamping lid from rotation to the open position.

(6) In another aspect, the disclosure, a planetary ball mill is disclosed. The planetary ball mill includes a carrier plate, a counterbalance system coupled to the carrier plate and a clamping system for clamping a grinding container to the carrier plate. The clamping system includes a base and a clamping lid including a clamping mechanism configured to secure a grinding container between the base and the clamping mechanism. The clamping lid is configured to pivot with respect to the base from a closed position to an open position, wherein in the open position the grinding container is removable from the clamping system. The system further includes a rotatable arm configured to pivot from a first position to a second position, wherein in the first position the rotatable arm prevents the clamping lid from rotation to the open position.

(7) In another aspect of the disclosure, a balancing system for a planetary ball mill is disclosed. The balancing system includes a first counterweight and a linkage system coupled between the first counterweight and a carrier plate of the planetary ball mill via a linkage system, the linkage system allows the first counterweight to move with respect to a rotational axis of the carrier plate.

(8) These and other aspects of the invention will become more fully understood upon a review of the detailed description, which follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is perspective view of an example planetary ball mill including a clamping system in a closed configuration, according to an aspect of the disclosure.
- (2) FIG. 2 is perspective view of the planetary ball mill of FIG. 1 with the clamping system in an open configuration, according to an aspect of the disclosure.
- (3) FIG. 3 is a side view of the clamping system of the planetary ball mill of FIGS. 1 and 2.
- (4) FIG. 4 is a front view of the clamping system of the planetary ball mill of FIGS. 1-3 engaged with a grinding container.
- (5) FIG. 5 is a top view of the clamping system of the planetary ball mill of FIGS. 1-4 engaged with a grinding container.
- (6) FIG. 6 is a perspective view of an example grinding container for use with the planetary ball mill of FIGS. 1-5.
- (7) FIG. 7A is a section view of the clamping system of the planetary ball mill of FIGS. 1-5 engaged with the grinding container of FIG. 6.
- (8) FIG. 7B is a view of one example of ball movement within the grinding container of the ball mill.
- (9) FIG. 8 is a rear perspective view of the ball mill of FIGS. 1-5.
- (10) FIG. 9A is a perspective view of an example grinding container design according to aspects of the disclosure.
- (11) FIG. 9B is front view of the example grinding container of FIG. 9A according to aspects of the disclosure.
- (12) FIG. 9C is a left side view of the grinding container of FIGS. 10A and 10B according to aspects of the disclosure.
- (13) FIG. 9D is a rear view of the grinding container of FIGS. 10A-10C according to aspects of the disclosure.
- (14) FIG. 9E is a right side view of the grinding container of FIGS. 9A-9D according to aspects of the disclosure.
- (15) FIG. 9F is a bottom view of the grinding container of FIGS. 9A-9E according to aspects of the disclosure;
- (16) FIG. 9G is a top view of the grinding container of FIGS. 9A-9F according to aspects of the disclosure.
- (17) FIG. 10A is a perspective view of an example grinding container lid design according to aspects of the disclosure.
- (18) FIG. 10B is top view of the example lid of FIG. 10A according to aspects of the disclosure.
- (19) FIG. 10C is a left side view of the lid of FIGS. 10A and 10B according to aspects of the disclosure.
- (20) FIG. 10D is a rear view of the lid of FIGS. 10A-10C according to aspects of the disclosure.
- (21) FIG. 10E is a right side view of the lid of FIGS. 10A-10D according to aspects of the disclosure.
- (22) FIG. 10F is a front view of the lid of FIGS. 10A-10E according to aspects of the disclosure.
- (23) FIG. 10G is a bottom view of the lid of FIGS. 10A-10F according to aspects of the disclosure.
- (24) FIG. 11A is a perspective view of an example grinding container bowl design according to aspects of the disclosure.
- (25) FIG. 11B is top view of the example bowl of FIG. 11A according to aspects of the disclosure.
- (26) FIG. 11C is a left side view of the bowl of FIGS. 11A and 11B according to aspects of the disclosure.
- (27) FIG. 11D is a rear view of the bowl of FIGS. 11A-11C according to aspects of the disclosure.
- (28) FIG. 11E is a right side view of the bowl of FIGS. 11A-11D according to aspects of the disclosure.

(29) FIG. 11F is a front view of the bowl of FIGS. 11A-11E according to aspects of the disclosure.
(30) FIG. 11G is a bottom view of the bowl of FIGS. 11A-11F according to aspects of the disclosure.

DETAILED DESCRIPTION

(31) The detailed description set forth below in connection with the appended drawings is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known components are shown in block diagram form in order to avoid obscuring such concepts.

I. TERMINOLOGY

(32) Throughout the disclosure, the terms substantially or approximately may be used as a modifier for a geometric relationship between elements or for the shape of an element or component. While the terms substantially or approximately are not limited to a specific variation and may cover any variation that is understood by one of ordinary skill in the art to be an acceptable variation, some examples are provided as follows. In one example, the term substantially or approximately may include a variation of less than 10% of the dimension of the object or component. In another example, the term substantially or approximately may include a variation of less than 5% of the object or component. If the term substantially or approximately is used to define the angular relationship of one element to another element, one non-limiting example of the term substantially or approximately may include a variation of 5 degrees or less. These examples are not intended to be limiting and may be increased or decreased based on the understanding of acceptable limits to one of skill in the relevant art.

(33) For purposes of the disclosure, directional terms are expressed generally with relation to a standard frame of reference when the system and apparatus described herein is installed in an in-use orientation. Further, in order to provide context to the current disclosure, a broad overview of the discovered deficiencies of various systems and an example implementation of the current disclosure and the advantages provided by the disclosure are described below. Further details of example implementations of the current disclosure are described in detail with reference to the figures below.

(34) Terms such as a, an, and the are not intended to refer to only a singular entity, but include the general class of which a specific example may be used for illustration. The terms a, an, and the may be used interchangeably with the term at least one. The phrases at least one of and comprises at least one of followed by a list refers to any one of the items in the list and any combination of two or more items in the list. All numerical ranges are inclusive of their endpoints and non-integral values between the endpoints unless otherwise stated.

(35) The terms first, second, third, and fourth, among other numeric values, may be used in this disclosure. It will be understood that, unless otherwise noted, those terms are used in their relative sense only. In particular, in some aspects certain components may be present in interchangeable and/or identical multiples (e.g., pairs). For these components, the designation of first, second, third, and/or fourth may be applied to the components merely as a matter of convenience in the description of one or more of the aspects of the disclosure.

(36) For context, an overview is provided of aspects of the disclosure and the advantages the disclosure provides. This overview, and the detailed description that follows, has been presented for purposes of illustration and description. It is not intended to be exhaustive nor to limit the disclosure to the forms described. Numerous modifications are possible in light of the above teachings, including a combination of the abovementioned aspects. Some of those modifications have been discussed and others will be understood by those skilled in the art. The various aspects were chosen and described in order to best illustrate the principles of the present disclosure and

various aspects as are suited to the particular use contemplated. The scope of the present disclosure is, of course, not limited to the examples or aspects set forth herein, but can be employed in any number of applications and equivalent devices by those of ordinary skill in the art. Rather, it is hereby intended the scope be defined by the claims appended hereto.

II. OVERVIEW

(37) As described in further detail below, aspects of the disclosure relate to an improved clamping system that simplifies the installation and removal—and securing of a grinding container **112** in to a ball mill **100**. Turning to FIGS. **1** and **2**, the clamping system **116** may include a base plate or ring **308** and a clamping lid **424** with a clamping member **344**. As shown in FIG. **2**, a support **324** connecting the base ring **308** may be configured to be pivoted to disengage from a boss **384** which allows the clamping lid **424** to be pivoted away from a grinding container **112**. Once the clamping lid **424** is pivoted away from the grinding container **112** the grinding container may be easily removed from the clamping system **116**. Similarly, a grinding container **112** may be easily installed and secured into the clamping system **116** by placing the grinding container into the clamping system and pivoting the clamping lid **424** into engagement with the lid of the grinding container **112**. Once the grinding container is in place, the support **324** may be pivoted upwards and into engagement with the boss **384** as shown in FIG. **1** and a clamp screw handle **348** may be rotated to cause the clamping member **344** to tighten down onto the lid of the grinding container **112**. The aforementioned system greatly increases the efficiency in securing and removing grinding containers **112** from a ball mill. Further, the aspects described below provide a secure connection between the ball mill and the grinding container.

(38) In another aspect described below, a balancing system is described. The balancing system allows a user to easily balance a ball mill by simply turning a knob. Additional aspects are described below.

III. DETAILED EXAMPLES

(39) FIG. **1** illustrates a partial perspective view of an example ball mill **100** with an example clamping system **116** according to aspects of the disclosure in a first, or closed, configuration. It is noted that external features (e.g., the outer casing) of the ball mill are omitted from FIGS. **1** and **2** to prevent obstruction of aspects of the disclosure. In one example, the planetary ball mill may be a commercially available benchtop planetary ball mill. For example, aspects of the disclosure may be usable with commercially available ball mill's such as the Retsch PM100, manufactured by Retsch of Germany; the Fritsch Pulverisette 6, manufactured by Fritsch of Germany, The Tencan Heavy-duty Full-directional Planetary Ball Mill, manufactured by Tencan of China, and/or planetary mills manufactured by Across International of New Jersey. Examples of ball mills having aspects that may be useable with the disclosure are described by Limb in U.S. Pat. No. 2,874,911, titled: Compound Movement Centrifugal Ball-Mill; Gamblin in U.S. Pat. No. 5,205,499, titled: Planetary Grinding Apparatus, the entirety of which are incorporated by reference herein.

(40) FIG. **2** is a perspective view of the example ball mill **100** with the example clamping system **116** in accordance with the disclosure in a second, or open configuration. The ball mill **100** and clamping system **116** may include a motor **104**, a gear train **108** (hidden from view in FIGS. **1** and **2**), a grinding container **112**, and a balancing system **120**.

(41) The motor **104** may be configured to drive the gear train **108**. The gear train **108** may for example be a planetary gear system. For example, an output shaft of the motor **104** may be coupled to a sun gear (not shown) of the gear train **108**. The sun gear may engage and drive one or more planetary gears (not shown) coupled to a ring gear. The ring gear may be coupled to or formed in a carrier plate **130**. The clamping system **116** and the balancing system **120** are coupled to the carrier plate **130** such that the clamping system **116** and the balancing system **120** rotate with the carrier plate **130** (e.g., about axis AA shown in FIG. **1**). Further, any one or combination of the planetary gears may cause rotation of the clamping system and/or grinding container clamped therein about axis BB shown in FIG. **1**. The clamping system **116** is configured to engage the grinding container

112 and secure the grinding container **112** during rotation. It is noted that while a planetary ball mill **100** is provided as an example, aspects of the current disclosure are likewise applicable to any system that requires or is improved by the secure clamping of a container or grinding container for application of a rotational and/or vibrating motion to the container clamped therein.

(42) FIG. 3 illustrates a side perspective view of the clamping system **116** without the grinding container **112** (FIGS. 1 and 2) clamped therein. FIG. 4 is a front perspective view of the clamping system **116** engaged with the grinding container **112**. FIG. 5 is a top view of the clamping system **116** engaged with the grinding container **112**. As shown in FIG. 1, the clamping system **116** includes a base member **302**, which includes a base plate **304** and a base ring **308**. The base plate **304** is coupled to the base ring **308**. A portion of the base plate **304** is received within an opening **132** (FIGS. 1-2) of the carrier plate **130** and is coupled to and/or includes a gear configured to engage the sun gear of the gear set **108** to facilitate rotation of the clamping system **116** (e.g., about axis BB in FIG. 1) as the carrier plate **130** rotates.

(43) The clamping system may further include a clamping lid **424** (FIG. 5), a first support arm **316**, a second support arm **320**, and a third support arm **324**. The first support arm **316**, second support arm **320**, and/or third support arm **324** may be connected to the base ring **308**. In one example, the first support arm **316** and second support arm **320** may be fixedly connected to the base ring **308** via one or more fasteners at first ends thereof (e.g., at respective first ends **352**, **356**). In one example, the first support arm **316** and/or second support arm **320** may be connected to the base ring **308** so as to not be rotatable with respect to the base ring **308**. For example, the first support arm **316** and/or second support arm may be connected to the base ring via two or more threaded fasteners that are configured to be threadably engaged with a respective plurality of threaded holes in the base ring **308**. One example of the aforementioned fastener configuration is shown as reference **351** in FIG. 3.

(44) Further, as shown in FIGS. 1, 2, and 4, the third support arm **324** may for example be pivotally connected to the base ring **308** at a first end **372** and configured to pivot or rotate about axis CC (FIG. 4) in a first direction DD (FIG. 2) and/or a second direction OO. The third support arm **324** may be configured to pivot from a first closed or clamped configuration (e.g., as shown in FIG. 1) to a second open configuration (e.g., as shown in FIG. 2). In one example implementation that allows the third support arm **324** to pivot about axis CC, the third support arm **324** may for example be connected to the base ring **308** via single threaded fastener (e.g., threaded fastener **325** in FIGS. 1, 2, and 4). It is noted that while a threaded fastener is shown, any configuration that allows the third support arm **324** to pivot may be implemented without departing from the scope of the disclosure (e.g., a bearing configuration, a bushing configuration, a slip fit rivet or other rotatable interface). Additional discussion relating to the operation of the clamping system is described in further detail below.

(45) The first support arm **316**, second support arm **320** and/or third support arm **324** may be connected to or connectable to a clamping lid configuration at second ends thereof (e.g., at respective second ends **360**, **364**, and/or **376**). Turning to FIG. 5, the clamping lid **424** may include a clamping body **328** which may include a first clamping body arm **332**, a second clamping body arm **336**, and a third clamping body arm **340**, a clamping member **344**, and a clamp screw handle **348**. The clamp screw handle **348** is configured to engage the clamping member **344** with a lid **134** of the grinding container **112**. In one example, the clamping body **328** may include a threaded portion configured to threadably engage with a threaded shaft (e.g., shaft **343** shown in FIGS. 3 and 4). The shaft **343** may have the clamping member **344** connected at a first end and the clamp screw handle **348** connected at the second end. In one example the clamp screw handle **348** may be fixed to the second end so that rotation of the clamp screw handle **348** causes rotation of the threads on the shaft **343** within the corresponding threads of the clamping body **328**, thus causing linear advancement of the clamping member **344**. For example, turning to FIG. 4, rotation of the clamp screw handle **348** in the direction EE denoted by the arrow causes the clamping member **344** to

advance in direction FF denoted by the arrow in FIG. 4. Thus, as described in further detail below, a user may rotate the clamp screw handle **348** to tighten the clamping member **344** against lid **134**. The aforementioned example may not only ensure that the grinding container **112** is securely mounted, but may also ensure that the lid **134** of the grinding container **112** is properly engaged and sealed with the bowl **138** of the grinding container **112**. In one example, the clamping member **344** may be rotatably connected to shaft via a bearing, bushing or other rotatable interface to allow a user to more easily rotate the clamp screw handle **348** once the clamping member **344** has made contact with the lid **134** during the tightening process.

(46) While one specific example is provided above, alternative methods of applying a downward force to the lid **134** of the grinding container **112** may be used without departing from the scope of the disclosure. For example, the clamping member **344** may be biased downward via a spring or other biasing member. In another example, the clamping member **344** may be connected to a spring cam mechanism that is configured to release spring pressure and thus apply a downward force to the lid **134** of the grinding container when the clamp screw handle is rotated or otherwise activated. Mechanisms for providing the aforementioned downward force, including the non-limiting example described above, may hereinafter be interchangeably referred to as a clamping mechanism.

(47) Turning to FIGS. 4 and 5, the support arms **316**, and **320** may be coupled between the base ring **308** and the clamping lid **424**. As mentioned above, a first end **352** of the first support arm **316** and a first end **356** of the second support arm **320** may be fixedly coupled to the base ring **308**. In one example, the first support arm **316** and/or second support arm **320** may be connected to the base ring **308** so as to not be rotatable with respect to the base ring **308**. For example, the first support arm **316** and/or second support arm may be connected to the base ring via two or more threaded fasteners that are configured to be threadably engaged with a respective plurality of threaded holes in the base ring **308**. One example of the aforementioned fastener configuration is shown as reference **351** in FIG. 3. A second end **360** of the first support arm **316** and a second end **364** of the second support arm **320** may be fixedly connected to a first pivot mount **333** and a second pivot mount **337**. The first pivot mount **333** and second pivot mount **337** may be configured to allow the clamping lid **424** to pivot with respect to the first support arm **316** and second support arm **320** about an axis PP (FIG. 5). For example, the first pivot mount **333** and second pivot mount **337** may be configured to allow for rotational movement of the second clamping body arm **336** and first clamping body arm **332** with respect to the first support arm **316** and second support arm **320** via hinges **364** and **368**. A first end **372** of the third arm **324** is rotatably coupled to the base ring **308**. A second end **376** of the third arm **324** includes a notch **380** that is configured to engage a boss **384** extending from the third clamping body arm **340** of the clamping lid **424**. The third arm **324** is movable between a first position (FIG. 1) in which the notch **380** is engaged with the boss **384** clamping lid **424** and a second position (FIG. 2) in which the notch **380** is disengaged from the boss **384** of the clamping lid **424**. In the first position, the position of the clamping lid **424** with respect to the base ring **308** is fixed so that the grinding container **112** may be secured between the clamping member **344** and the base plate **304**. When the third arm **424** is in the first position, the clamp screw handle **348** may also be actuated (e.g., rotated) to secure the clamping member **344** to the grinding container lid **134**, thereby preventing the grinding container **112** from opening or otherwise moving undesirable during rotation of the ball mill. When the third arm **324** is in the second position, the clamping lid **424** may be pivoted toward or away from the grinding container **112** allowing for the grinding container **112** to be easily removed. It is noted that while a combination of pivoting downward of the clamping lid **424**, engagement of the third support arm **324** with the boss **384** and the actuation of the clamp screw handle **348** thereafter is described above, in one example the clamping member **344** may be configured to automatically apply a downward force to the container lid once clamping lid **424** is pivoted downward. For example, the clamping member **344** may be downwardly biased via a spring or other biasing member so that

pressing downward on the clamping lid **424** causes a downward force to be applied to the lid **134** of grinding container **112**.

(48) FIG. **6** illustrates an example of a grinding container **112** usable with a clamping mechanism (e.g., clamping system **116** described above). As mentioned above, one non-limiting example use of the clamping mechanism and/or grinding container is with a planetary ball mill. The grinding container **112** may include a lid **134** and a bowl **138**. The lid **116** may include protrusion **142** configured to engage with a concave portion of the clamping member **344** of. In one example, the lid may include a substantially flat surface **143** with the protrusion **142** including an upper surface **142b** and a beveled or angled second surface **142a**. The upper surface **142b** and beveled or angled surface **142a** may be configured to fit within or otherwise engage with similarly dimensioned beveled or angled second surface **442a** (FIG. **2**) and/or an upper surface **442b** of the clamping member **344**. FIG. **7** illustrates a section view of the grinding container **112** engaged with the clamping system **116**. The aforementioned protrusion of the grinding container **112** and matching concave portion of clamping member **344** may ensure that the container **112** and/or lid **134** is properly centered when the container is clamped within the clamping system **116**. Further, the aforementioned protrusion of the grinding container **112** and matching concave portion of clamping member **344** may ensure that the lid **134** is properly sealed with respect to the bowl **138** when the container is secured within the clamping system **116**. The lid **134** of the grinding container **112** may further include a sealing member such as an O-ring **144** (FIG. **7**). The O-ring may for example be replaceable to allow the grinding container **112** to be reused or refurbished if the seal **144** is damaged or worn. The container may **112** may further include an alignment member on the bottom thereof. For example as shown in FIG. **9F**, the bottom of the grinding container may include a concave portion **405** configured to receive or otherwise engage with an alignment pin **305** (FIG. **3**) of the base plate **304**. The engagement between the concave portion **405** and the alignment pin **305** may further assist with the alignment of the container within the clamping system **116** and may further ensure that the container does not loosen or otherwise move with respect to the clamping system **116**.

(49) FIG. **7** shows an example of the operation of the grinding container **112**. As shown in FIG. **7**, an amount of material to be ground **700** and a plurality of grinding balls **704** may be placed into the bowl **138**. In operation, grinding container **112** rotates about the axis A and/or about axis AA (FIG. **1**). The arrows in FIG. **7B** indicate movement of the grinding balls **704** and the ball mill during rotation. During rotation, centrifugal motion causes the grinding balls **704** to travel along the perimeter of the bowl **138**. The balls **704** travel around the perimeter of the jar until they reach a critical point when they fling across to the opposite wall of the jar as indicated by the broken arrows in FIG. **7B**. The movement of the balls **704** through the material **700** results in both shearing and impact forces. The impact of the balls **704** crushes larger chunks of the material to be ground and the shearing grinds the material to be ground to a fine powder. During the process of grinding the material **700**, the Balls **704** may be replaced with balls having different dimensions and/or mass. For example, a user of the ball mill may wish to mill or grind the material to be ground **700** to a very small (nano) particle size, the user may first grind the sample using a first size of balls, and then grind the sample using a second size of balls that are smaller than the first size of balls.

(50) Referring now to FIGS. **1**, **2**, and **8**, in one aspect of the disclosure a balancing system is disclosed. The balancing system may for example be used in conjunction with the aspects described above and/or may be used separately to provide a system or method for balancing a planetary ball mill. One example of a ball mill **100** with the example clamping system balancing system **116** is usable with the ball mill that includes a motor **104**, a gear train **108** (hidden from view in FIGS. **1** and **2**), a grinding container **112**. The motor **104** is configured to drive the gear train **108**. The gear train **108** may for example be a planetary gear system.

(51) For example, an output shaft of the motor **104** may be coupled to a sun gear (not shown) of the gear train **108**. The sun gear may engage and drive one or more planetary gears (not shown)

coupled to a ring gear. The ring gear may be coupled to or formed in a carrier plate **130**. A clamping system **116** configured to secure a grinding container and a balancing system **120** are coupled to the carrier plate **130** such that the clamping system **116** and the balancing system **120** rotate with the carrier plate **130** (e.g., about axis AA shown in FIG. 1). Further, any one or combination of the planetary gears may cause rotation of the clamping system and/or grinding container clamped therein about axis BB shown in FIG. 1. During the aforementioned rotation, performance of the ball mill may be improved by providing a counterweight to prevent excessive vibration or “walking” of the planetary ball mill. Further, because the ball mill is useable with different media, balancing of the ball mill requires adjustment due to different weights of media that may be placed in the grinding container. In one aspect an adjustable balancing system is disclosed.

(52) The balancing system **120** includes a first counterweight **804**, a first linkage **808**, a second counterweight **812**, and a second linkage **816**. The first counterweight **804** may be coupled to the carrier plate **130** via the first linkage **808**. The second counterweight **812** may be coupled to the carrier plate **130** via the second linkage **816**. As the carrier plate rotates **130**, the first and second counterweights **804** and **812** counterbalance the movement of the grinding container **112** and the contents therein during rotation. The first and second counterweights may be configured to be adjustable toward and away from the central rotational axis of the carrier plate **130**. For example, The balancing system **120** may include a knob **820** that may be actuated (e.g., rotated) to change a distance that the first and second counterweights **804**, **812** are relative to the center of the carrier plate **130** during rotation as denoted by the arrows in FIG. 8. For example, when the load in the grinding container **112** is heavy, an operator may actuate (e.g., rotate) the knob **820** to increase the distance that the first and second counterweights **804**, **812** are relative to the center of the carrier plate **130**. For example, when the load in the grinding container **112** is light, an operator may actuate the knob **820** to decrease the distance that the first and second counterweights **804**, **812** are relative to the center of the carrier plate **130** during rotation. When the load in the grinding container **112** is high, the operator may actuate the knob to increase the distance that the first and second counterweights **804** and **812** are relative to the center of the carrier plate.

(53) In one example, the first linkage **816** may be connected at a first end to a sliding mount **823**, the sliding mount **823** may for example be configured to move linearly toward and away from the center of rotation of the carrier plate **130** via a track **821** when the knob **820** is rotated in a first direction or a second direction. For example, the knob **820** may be connected to a threaded shaft **822** that is configured to threadingly engage with a threaded portion of the sliding mount **823**. Turning the knob **820** causes the threads of the threaded shaft **822** to advance and causes linear motion of the sliding mount **823** along track **821**. Both the first linkage **808** and the second linkage **816** may be pivotally connected to the sliding mount **823** at a first end at carrier plate pivots **809** and **817** at a second end. Thus, as the sliding mount moves away from or towards the central axis AA of the carrier plate **130** the counterweights **804** and **812** move both towards or away from the central axis AA and closer to and further away from one another which further assists with the stabilization of the carrier plate **130** during ball mill operation. The aforementioned configuration allows a user to efficiently balance the carrier plate **130** of the ball mill. In one example, additional counterweights may be added to balance a larger container, multiple containers, and/or heavier sample and/or ball weights. For example, additional weights may be added to the top of each one of counterweights **804** and/or **812**.

(54) FIGS. 9A-9G show an example grinding container design usable with aspects of the disclosure. In one example implementation the grinding container **912** may be analogous to grinding container **112** described above.

(55) FIGS. 10A-10G show views of a lid design usable with aspects of the disclosure. In one example implementation the lid **934** may be analogous with grinding container lid **134** described above.

(56) FIGS. 11A-11G show views of an example bowl design usable with aspects of the disclosure. In one example implementation the bowl 938 may be analogous with bowl 138 described and above.

(57) It is noted that any one or a combination of the components mentioned above may be provided as a kit and may come pre-packaged as a kit or system with instructions for use.

(58) This written description uses examples to disclose aspects of the invention, including the preferred embodiments, and also to enable any person skilled in the art to practice the aspects thereof, including making and using any devices or systems and performing any incorporated methods. The patentable scope of these aspects is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims. Aspects from the various embodiments described, as well as other known equivalents for each such aspect, can be mixed and matched by one of ordinary skill in the art to construct additional embodiments and techniques in accordance with principles of this application.

Claims

1. A clamping system for a planetary ball mill, the clamping system comprising: a base; a plurality of support arms connected to the base; a clamping lid connected to the plurality of support arms and including a clamping mechanism configured to clamp a grinding container between the base and the clamping mechanism, wherein the clamping lid is pivotally coupled to the plurality of support arms and is configured to pivot with respect to the base from a closed position to an open position, wherein in the open position the grinding container is removable from the clamping system; and a rotatable arm connected to the base and configured to pivot from a first position to a second position, wherein in the first position the rotatable arm prevents the clamping lid from rotating to the open position.
2. The clamping system of claim 1, wherein when the rotatable arm is in the second position, the clamping lid is rotatable to the open position.
3. The clamping system of claim 1, wherein the plurality of support arms are fixed arms are fixedly coupled to the base.
4. The clamping system of claim 1, wherein the rotatable arm includes a notch configured to engage a boss of the clamping lid.
5. The clamping system of claim 1, wherein in the closed position, the clamping lid engages the grinding container and in the open position the clamping lid is spaced from the grinding container.
6. The clamping system of claim 5, wherein when the clamping lid is in the closed position, the clamping mechanism is configured to bias a lid of the grinding container towards the base of the clamping system to provide engagement between the lid and a bowl of the grinding container.
7. A clamping system for a planetary ball mill, the clamping system comprising: a base; a plurality of support arms connected to the base; a clamping lid connected to the plurality of support arms and including a clamping mechanism configured to clamp a grinding container between the base and the clamping mechanism, wherein the clamping lid is configured to pivot with respect to the base from a closed position to an open position, wherein in the open position the grinding container is removable from the clamping system; and a rotatable arm connected to the base and configured to pivot from a first position to a second position, wherein in the first position the rotatable arm prevents the clamping lid from rotating to the open position, wherein in the closed position, the clamping lid engages the grinding container and in the open position the clamping lid is spaced from the grinding container, wherein when the clamping lid is in the closed position, the clamping mechanism is configured bias a lid of the grinding container towards the base of the clamping

system to provide engagement between the lid and a bowl of the grinding container, and wherein the clamping mechanism comprises a threaded rod that is configured to be rotated causing the clamping mechanism to tighten against the lid of the grinding container.

8. The clamping system of claim 1, wherein the clamping mechanism further comprises a clamping member with an engagement member with at least one surface configured to engage with a corresponding surface of an engagement member on a lid of the grinding container.
